MIL-D-83934A(USAF)
21 September 1972
SUPERSEDING
MIL-D-83934 (USAF)
31 July 1970

MILITARY SPECIFICATION

DRYER, AIR, LOW PRESSURE

1. SCOPE

1.1 Scope. This specification covers air dryers for use in low pressure cable systems. The dryer shall remove particulate matter, oil aerosol, and water vapor from influent atmospheric air and furnish clean, dry air as specified herein. The unit shall be automatic in operation. It is intended that the parameters given herein be representative of commercially available air dryers.

1.2 Classification.

1.2.1 Types. Air dryers shall be the following types as specified (see 6.2). Inlet air shall be 95° F db and 80° F wb at 14.7 psia pressure.

Type I - Maximum moisture content of effluent air at 14.7 psia and 70° F shall be 310 PPM by weight (equivalent to - 18° F dewpoint or 2% R.H. at 70° F).

Type II - Maximum moisture content of effluent air at 14.7 psia and 70°F shall be 80 PPM by weight (equivalent to - 40° F dewpoint).

1.2.2 Capacity. The minimum capacity (effluent air flow rate) expressed as standard cubic feet per day (SCFD) shall be as specified in para 6.2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated, Plywood

Military

MIL-P-116	Preservation, Methods of

FSC 4440

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MIL-D-3716	Desiccants, Activated, for Dynamic Dehumidification
MIL-D-22577	Heating Elements, Electrical, Cartridge, Strip and Tubular Type

STANDARDS

Military

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Fire Protection Association

Pamphlet 70 National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110).

American Society of Mechanical Engineers (ASME)

Power Test Codes, Unfired Pressure Vessels Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N. Y. 10017).

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 Design.

3.2.1 Construction. The air dryer shall be a complete unit consisting of a drive motor, compressor, drying system, and other components necessary to provide a fully automatic unit. The unit shall be suitable for continuous or intermittent operation at conditions specified herein. The air dryer shall be housed in a metal enclosure constructed as a single cabinet unit. The maximum size of the unit shall be as specified in para 6.2. When specified (see 6.2) the air dryer shall be contained within a weatherproof enclosure suitable for pole mounting. When the dryer design incorporates a

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combination of refrigerant and desiccant systems, the individual design paragraphs of this specification shall apply.

3.2.2 Design (refrigeration type). When compression-refrigeration units are furnished they shall be the hermetically sealed type and use refrigerant 12 or 22.

3.2.3 Design (desiccant type). When compression desiccant units are furnished each tower shall have at least one removable head to facilitate replacement of desiccant and allow inspection of tower interior. Desiccant shall conform to the requirements of MIL-D-3716.

3.2.4 Heaters. When heaters are used in the desiccant regenerating cycle, heater elements shall be in accordance with MIL-H-22577.

3.2.5 Storage tank. Air storage tanks or desiccant towers shall be constructed of corrosion resistant material and shall comply with the requirements of ASME code for unfired pressure vessels.

3.2.6 Regulating valve. The air discharge line shall include a pressure regulating valve suitable for setting the outlet pressure at any point between 0 and the maximum design pressure as specified in para 6.2.

3.2.7 Safety valve. The system shall include a pressure relief valve to insure against excessive pressure which may induce a personnel hazard.

3.2.8 Pressure gage. Pressure gages shall monitor both the effluent air pressure and the internal system pressure.

3.2.9 Controls. Controls shall include a manual circuit breaker type on-off switch and the control knob for the pressure regulator. Gage and controls shall be weather protected if the unit is to be pole mounted (see 6.2).

3.2.10 Materials. All air and water lines shall be constructed of corrosion resisting material.

3.2.11 Humidity alarm. A humidity alarm shall be provided to monitor the effluent air. If the moisture content of the dry air rises above the value specified the relay will close to give a signal (to the connected alarm system).

3.2.12 Interchangeability. All parts having the same manufacturer's part number shall be interchangeable.

3.2.13 Electrical. The dehydrator shall operate on 115, 208, 220 or 230 volt, single phase, 60 hertz power as specified para 6.2. All components and wiring shall comply with requirements of the National Electrical Code.

3.2.14 Air inlet filter. An air inlet filter shall be provided to prevent insects, dust, or other foreign materials from entering the system.

3.2.15 Finishes and protective coatings.

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3.2.15.1 Cleaning, painting, plating, anodic films, and chemical treatments. All cleaning, painting, plating, chemical films, and chemical treatment shall be in accordance with MIL-STD-808, Type II.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the air dryer shall be classified as follows:

a. Preproduction tests - see 4.3.
b. Acceptance tests - see 4.4.

4.3 Preproduction tests. One air dryer representative of the finished product in all respects shall be subjected to the tests specified in para 4.5.

4.4 Individual tests. Each unit to be offered for acceptance shall be subjected to the tests of para 4.5.1, 4.5.2.2.

4.5 Test method.

4.5.1 Examination of product. The air dryer shall be inspected to insure compliance with the requirements of this specification.

4.5.2 Operating test. The tests specified in 4.5.2.1 shall be conducted to verify satisfactory performance. Dewpoint and flow rate measurements of influent and effluent air shall be recorded at intervals which are adequate to determine performance.

4.5.2.1 Endurance test. The air dryer shall be operated at ambient air conditions of 95° F dry bulb and 80° F wet bulb or at ambient specified in para 6.2g. The unit shall be operated continuously at this condition for a time period of 500 hours. Throughout this test the effluent air dewpoint shall not be higher than specified in para 1.2. The air dryer shall deliver the volume specified in para 6.2 at the design pressure as specified in para 6.2.

4.5.2.2 Individual test. Each unit shall be operated for a period of at least 8 hours before shipment. Throughout this test the effluent air, volume, pressure, and dew point shall be as specified in para 6.2.

4.5.2.3 Humidity alarm. Operation of the humidity alarm shall be demonstrated by increasing the R. H. sufficiently to activate the alarm system.

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4.5.3 Maintenance. Not more than 1 man hour shall be consumed for maintenance during all of the preproduction test.

4.5.4 Rejection and retest. When an air dryer fails to meet the specification, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and corrected.

4.7 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully correct all defects.

4.8 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 General. The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Levels of preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A. Air dryers shall be preserved in accordance with MIL-P-116, Method II. Unit quantities shall be one each.

5.2.2 Level C. Air dryers shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.3 Levels of packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. Air dryers preserved and packaged to meet 5.2.1 shall be packed in exterior type shipping containers meeting PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.3.2 Level B. Air dryers preserved and packaged to meet 5.2.1 shall be packed in domestic type exterior containers meeting PPP-B-601. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with applicable container specification or appendix thereto.

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5.3.3 Level C. Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.4 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

AIR DRYER

In addition, the following marking information shall be included:

LABEL, "HANDLE WITH CARE"

AND IDENTIFY "UP" POSITION

6. NOTES

6.1 Intended use. To provide low pressure dry air for the purpose of pressurizing cable systems or radar waveguide systems.

6.1.1 Suggestion for selection. The type II dryer will produce a lower dew point and is best suited where frost cannot be tolerated inside a pressurized system if the ambient temperature drops below the dew point. Frost inside of a waveguide will lower the quality of the signal and cause arcing. Therefore, a radar or microwave system located in extremely cold climates with prolonged temperatures of less than -15° F should have a type II air dryer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type air dryer required (see 1.2.1).
c. Capacity (SCFD) required (see 1.2.2).
d. Pressure (psig) required (see 3.2.6).
e. Size limitations (height, length, width) (see 3.2.1).
f. When a weatherproof cabinet suitable for pole mounting is required (see 3.2.1).
g. Endurance test ambient condition, if different (see 4.5.2.1).
h. Operating voltage required (see 3.2.13).
i. Level of preservation, packaging and packing required (see 5.2 & 5.3).

CUSTODIAN:
Air Force - 84

PREPARING ACTIVITY:
Air Force - 84

PROJECT NUMBER 4440-F025

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
MIL-D-83924A(USAF) Dryer, Air, Low Pressure

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. 47 AFLC RAFB,GA JUN 67 - 500

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